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Protection sheet, press through package

Abstract

A protection sheet is to be attached to a press through package that includes a sheet-shaped container including storing portions and a cover that seals the storing portions. The protection sheet includes a resin layer and a first adhesive layer. The resin layer is releasable from the press through package and includes slits from which the resin layer starts to be released. The first adhesive layer is bonded to the resin layer such that a surface of the first adhesive layer that is an opposite side from the resin layer is to be bonded to the cover.

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Background/Summary

(1) This application is a national phase of International Application No. PCT/JP2022/023080 filed Jun. 8, 2022, which claims priority to Japanese Application No. 2021-095632 filed Jun. 8, 2021, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present technology relates to a protection sheet and a press through package.

BACKGROUND ART

(3) Conventionally, a press through package (PTP) has been known as a packaging material that includes storing portions storing objects such as medical tablets. Child-resistant functions have been recently proposed for the PTP such that the packaging material is not opened by children by mistake. One example is disclosed in Patent Document 1. A PTP (PTP packaging body) described in Patent Document 1 includes a base layer and a cover. The base layer includes an aluminum foil that can be pressed and broken. The cover includes a cover film that is disposed on the base layer so as to be releasable from the base layer. A non-adhesive section where no adhesive treatment is provided is formed on an interface portion between the base layer and the cover film in an edge portion of the cover. In opening the PTP, the portion of the cover film including the non-adhesive section is used as a holding portion such that children cannot open the PTP easily.

PRIOR ART DOCUMENT

Patent Document

(4) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2005-82179

Problem to be Solved by the Invention

(5) However, in the PTP described in Patent Document 1, the cover has a complicated layered structure and therefore, the producing process also becomes complicated. Releasing may be caused on the interface between other layers (for example, between the aluminum foil of the base layer and adjacent layer) other than the bonded portion where the holding portion of the cover film and the base layer are bonded. The cover may not be released stably.

SUMMARY OF THE INVENTION

(6) The present technology was made in view of the above

(7) circumstances. An object is to achieve a child-resistant function with a simple configuration.

Means for Solving the Problem

(8) Means for solving the problem is described below.

(9) <1> A protection sheet is to be attached to a press through package that includes a sheet-shaped container including storing portions and a cover that seals the storing portions. The protection sheet includes a resin layer and a first adhesive layer. The resin layer is releasable from the press through package and includes slits from which the resin layer starts to be released. The first adhesive layer is bonded to the resin layer such that a surface of the first adhesive layer that is an opposite side from the resin layer is to be bonded to the cover.

(10) <2> In the protection sheet according to <1>, every two slits of the slits are provided for each of the storing portions so as to be opposite each other and sandwich a portion of the resin layer overlapping a corresponding storing portion in a plan view, and each of the slits has an elongated

shape and extends from a portion of an outer edge of the resin layer in a direction that crosses the outer edge.

(11) <3> In the protection sheet according to <2>, a cut portion is on a surface of the resin layer opposite the first adhesive layer and the cut portion extends to connect extending ends of the every two slits.

(12) <4> In the protection sheet according to <2> or <3>, the resin layer has a rectangular planar shape as a whole, and each of the slits extends from a portion of a long side included in the outer edge along a short-side direction.

(13) <5> In the protection sheet according to <2> or <3>, the resin layer has a rectangular planar shape as a whole, and each of the slits extends from a portion of a short side included in the outer edge along a long-side direction.

(14) <6> In the protection sheet according to any one of <1> to <5>, the first adhesive layer is an adhesive agent or an adhesive tape that includes acryl-based material, urethane-based material, silicon-based material, or a rubber-based material.

(15) <7> The protection sheet according to any one of <1> to <6> further includes a metal layer disposed on an opposite side from the first adhesive layer with respect to the resin layer, and an IC mounted on the metal layer to overlap at least a portion of one of the slits in a plan view, the IC being for communicating with an external device.

(16) <8> The protection sheet according to any one of <1> to <7> further includes a release paper that is disposed on an opposite side from the resin layer with respect to the first adhesive layer and to be released from the first adhesive layer before being attached to the cover such that the first adhesive layer is uncovered.

(17) <9> A press through package includes the protection sheet according to any one of <1> to <7>.

(18) <10> the press through package according to <9>, the protection sheet has a planar shape and a planar size that are substantially same as those of the sheet-shaped container and the cover.

Effects of the Invention

(19) According to the present technology, a child-resistant function can be achieved with a simple configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an exploded perspective view of a PTP according a first embodiment.

(2) FIG. 2 is a plan view of a PTP body portion.

(3) FIG. 3 is a cross-sectional view taken along A-A line in FIG. 2.

(4) FIG. 4 is a plan view of a protection sheet.

(5) FIG. 5 is a cross-sectional view taken along B-B line in FIG. 4.

(6) FIG. 6 is a cross-sectional view of the PTP taken along a position of B-B line in FIG. 4.

(7) FIG. 7 is a perspective view of the PTP from which the protection sheet is released.

(8) FIG. 8 is an upper view of a PTP according to a second embodiment.

(9) FIG. 9 is a cross-sectional view taken along C-C line in FIG. 8.

(10) FIG. 10 is a plan view of a PTP body portion according to another embodiment.

(11) FIG. 11 is a plan view of a PTP body portion according to a further embodiment.

(12) FIG. 12 is a plan view of a protection sheet according to the other embodiment.

(13) FIG. 13 a plan view of a protection sheet according to the further embodiment.

MODES FOR CARRYING OUT THE INVENTION

First Embodiment

(14) A first embodiment will be described with reference to FIGS. 1 to 7. In this embodiment

section, a protection sheet **30A** and a press through package **10** (PTP, one example of a packaging material) to which the protection sheet **30A** is attached will be described. An X-axis, a Y-axis and a Z-axis may be present in each of the drawings and each of the axial directions represents a direction represented in each drawing.

(15) As illustrated in FIG. 1, the PTP **10** includes a PTP

(16) body portion **11** that stores objects T (medical tablets) and a protection sheet **30A** that is attached to the PTP body portion **11**. As illustrated in FIGS. 2 and 3, the PTP body portion **11** includes a sheet-shaped container **13** and a cover **14**. The sheet-shaped container **13** includes storing portions **12** in which the objects T are stored. The cover **14** is attached to cover openings **12A** of the storing portions **12** and the storing portions **12** are sealed. The PTP body portion **11** has a parallelepiped (rectangular) plan view shape and the protection sheet **30A** also has a plan view shape similar to that of the PTP body portion **11**. The protection sheet **30A** has a plan view size such that the whole PTP body portion **11** can be covered. In this embodiment, the protection sheet **30A** and the PTP body portion **11** have a substantially same plan view size.

(17) As illustrated in FIGS. 2 and 3, the sheet-shaped container **13** includes the storing portions **12** that are recessed (project in a semispherical shape) to be away from the cover **14**. The sheet-shaped container **13** is formed by processing a resin sheet having a thickness of from about 60 μm to 400 μm into a predefined shape. The sheet-shaped container **13** is a known resin container for a PTP and made of transparent thermoplastic resin such as polyvinyl chloride, polyvinylidene chloride, polypropylene, and amorphous polyethylene terephthalate. Each storing portion **12** has a circular plan view shape following the shape of the object T. In this embodiment, five storing portions **12** are arranged at predefined intervals in an X-axis direction (a long side direction) and two storing portions **12** are arranged at predefined intervals in a Y-axis direction (a short side direction). A total of ten storing portions **12** are formed.

(18) The cover **14** is a flat sheet member and made of thin material that can be pressed to release the object. Portions of the cover **14** other than the portions covering the openings **12A** of the storing portions **12** are bonded to the sheet-shaped container **13** as illustrated in FIG. 3. The cover **14** includes a heat sealing adhesive layer **15**, an aluminum foil **16**, and an overcoat layer **18** that are disposed on top of each other from the sheet-shaped container **13** side. The cover **14** is bonded to the sheet-shaped container **13** with the heat sealing adhesive layer **15** such that the storing portions **12** are sealed. The heat sealing adhesive layer **15** is a known thermal agent for a PTP. For example, polyester-based adhesive, polypropylene-based adhesive, and vinyl chloride-based adhesive may be used. The overcoat layer **18** is a protection layer made of transparent resin and protects the aluminum foil **16**.

(19) As illustrated in FIG. 1, the protection sheet **30A** is attached to a sheet surface of the cover **14** of the PTP body portion **11**. As illustrated in FIG. 5, a release paper **50** is attached to a surface of a low adhesive layer **38** (one example of a first adhesive layer) that includes a surface configured as a lower surface of the protection sheet **30A** before the protection sheet **30A** is attached to the cover **14**. The low adhesive layer **38** is uncovered by releasing the release paper **50** from the low adhesive layer **38**. As illustrated in FIG. 6, the uncovered low adhesive layer **38** is attached to the cover **14**. In this description, the opening detection sheet from which the release paper **50** is released is represented by a reference sign of **30A** and the opening detection sheet before the release paper **50** is released is represented by a reference sign of **30**. With the release paper **50**, the opening detection sheet **30** can be stored and transferred as a single object (a separate object) before being attached to the PTP body portion **11**.

(20) As illustrated in FIGS. 5 and 6, the protection sheet **30A** is a layered member including a resin layer **36** and the low adhesive layer **38**. As illustrated in FIG. 6, the resin layer **36** and the cover **14** of the PTP body portion **11** are bonded to each other with the low adhesive layer **38**. The resin layer **36** and the release paper **50** are releasably bonded to each other with the low adhesive layer **38** before the protection sheet **30A** is bonded to the PTP body portion **11**. The low adhesive layer **38** is

adhesive agent or an adhesive tape made of acryl-based material, urethane-based material, silicon-based material, or rubber-based material. The resin layer **36** is a resin sheet that can be bonded to the low adhesive layer **38** and is made of resin such as polyethylene terephthalate (PET), polystyrene, polyurethane, polypropylene, and polyimide. The resin layer **36** preferably has a predefined thickness or more so as to be easily held by a user when the resin layer **36** is released. (21) As illustrated in FIGS. **4** and **6**, the resin layer **36** and the low adhesive layer **38** include first slits **37** and half cutting portions **39**. The first slits **37** are through the resin layer **36** and the low adhesive layer **38**; however, the first slit **37** may necessarily be through at least the resin layer **36**. The resin layer **36** has a rectangular plan view shape having long sides and short sides and includes an outer edge **36A**. Each first slit **37** extends from the long side of the outer edge **36A** along a short side direction (a direction crossing the long side of the outer edge **36A**). Each first slit **37** extends in a thin elongated shape (specifically, in an I-shape). One end **37A** of each first slit **37** is at the outer edge **36A** of the resin layer **36**. With such a configuration, a user can use the one end **37A** as a starting point of releasing and release the resin layer **36** from the one end **37A** side along the first slit **37**. Two first slits **37** are provided for each of the storing portions **12** in a surface area of the resin layer **36**. Every two first slits **37** are disposed to be opposite each other so as to sandwich a portion of the resin layer **36** that overlaps the outer edge (an opening edge of the opening **12A**) of the storing portion **12**. With such a configuration, as illustrated in FIG. **7**, a user can easily peel and release the protection sheet **30A** that covers a desired one of the storing portions **12** to be opened. (22) As illustrated in FIG. **4**, a cut portion **39** (so-called half cutting portion) is formed in the resin layer **36** and the low adhesive layer **38**. The cut portion **39** extends to connect extending ends **37B** (another end) of the respective two first slits **37** that are provided for every storing portion **12**. The half cutting portion **39** is formed by cutting through the low adhesive layer **38** in an entire thickness and cutting the resin layer **36** in a half thickness from a low adhesive layer **38** side surface.

Accordingly, as illustrated in FIG. **7**, when the protection sheet **30A** that covers the desired storing portion **12** to be opened is peeled and released, the released portion is easily bent along the half cutting portion **39** and the cover **14** is easily kept in an uncovered state.

(23) According to the PTP **10** previously described, a user releases the protection sheet **30A** that covers the desired storing portion **12** and then, presses the object **T** in the desired storing portion **12** toward the cover **14** to break the cover **14**. Thus, the PTP **10** can be opened. To open the PTP **10**, two operations of peeling and pushing are necessary and therefore, children are less likely to open the PTP **10** by mistake and the child-resistant function can be achieved. The operation of peeling that is additionally performed to achieve the child-resistant function can be performed easily with the first slits **37** and the half cutting portion **39** and therefore, it is not a troublesome operation for a user. The PTP **10** is a packaging material that cannot be easily opened by children but can be opened by a user (an adult) without difficulty.

(24) The protection sheet **30A** has a layered structure that is not complicated and can be widely used for (attached to) the PTP body portion **11**, which is generally used. Therefore, the process of producing the protection sheet **30A** and the PTP **10** is not complicated. With the protection sheet **30A**, the child-resistance function can be achieved with a simple configuration.

Second Embodiment

(25) A protection sheet **130A** according to a second embodiment and a PTP **110** to which the protection sheet **130A** is attached will be described with reference to FIGS. **8** and **9**. The protection sheet **130A** differs from the first embodiment in the following points. The protection sheet **130A** includes an upper high adhesive layer **34A**, a base layer **35**, a lower high adhesive layer **34B**, and a metal layer **32** that are disposed on top of each other on the resin layer **36** and ICs (integrated circuits) **40** that are for communicating with an external device are mounted on the metal layer **32**. In the second embodiment, configurations, operations, and effects same as those of the first embodiment will not be described.

(26) As illustrated in FIG. **9**, the protection sheet **130A** includes the low adhesive layer **38**, the resin

layer **36**, the lower high adhesive layer **34B**, the base layer **35**, the upper high adhesive layer **34A**, and the metal layer **32** that are disposed on top of each other in this sequence from the lower side (PTP body portion **11** side). The metal layer **32** is an aluminum layer, for example, and is an antenna wire layer including a wire that is configured as an antenna. The antenna wire may be formed in a unitary form in an entire area of the metal layer **32** or multiple antenna wires may be formed in divided areas of the metal layer **32**, respectively. With the antenna wires being formed in the respective divided areas, as illustrated with one dot chain lines in FIG. **8**, the area of the metal layer **32** may be divided into rectangular sections corresponding to the respective storing portions **12** and the antenna wire may be formed in each of the divided sections (for example, ten sections in this embodiment).

(27) The PTP **110** is configured to perform wireless communication with an external communication device via the metal layer **32** (the antenna wire layer) and the ICs **40**. The ICs **40** and the antenna included in the metal layer **32** are configured to receive a signal transmitted by the external communication device and send a signal to the external communication device in response to the receiving of the signal. A near field communication technology such as radio-frequency identification may be used as a wireless communication method; however, the communication method is not particularly limited.

(28) As illustrated in FIG. **8**, the metal layer **32** includes second slits **33** (ten second slits **33** in this embodiment) having a thin elongated plan view shape (specifically an I-shape). One second slit **33** is formed for each storing portion **12**. As illustrated in FIGS. **8** and **9**, the second slit **33** overlaps one of the two first slits **37**, which are provided for every storing portion **12**, in a plan view. Each second slit **33** extends from an outer edge **32A** of the metal layer **32** in a form of an I-shape. Each second slit **33** extends through the metal layer **32** and the upper high adhesive layer **34A**. In this embodiment, each second slit **33** extends in the I-shape until it overlaps the half cutting portion **39**. However, the second slit **33** may not necessarily extend to the half cutting portion **39** as long as the second slit **33** overlaps a portion of the first slit **37** in a plan view.

(29) As illustrated in FIG. **8**, the base layer **35** is bonded to the metal layer **32** with the upper high adhesive layer **34A** and supports the metal layer **32**. With the base layer **35**, shape stability of the metal layer **32** and heat resistant property of the metal layer **32** during the manufacturing processing can be improved. The base layer **35** is made of resin material having insulating property. The thickness of the base layer **35** is (for example, from about 25 μm to 50 μm) much smaller than that of the resin layer **36**. Specifically, a PET (polyethylene terephthalate) film may be used as the base layer **35**.

(30) As illustrated in FIG. **9**, the upper high adhesive layer **34A** is between the metal layer **32** and the base layer **35** to bond the metal layer **32** and the base layer **35** firmly. The lower high adhesive layer **34B** is between the base layer **35** and the resin layer **36** to bond the base layer **35** and the resin layer **36** firmly. The upper high adhesive layer **34A** and the lower high adhesive layer **34B** are adhesive agent or adhesive tapes made of acryl-based material, urethane-based material, silicon-based material, or rubber-based material. The upper high adhesive layer **34A** and the lower high adhesive layer **34B** have adhesive force greater than that of the low adhesive layer **38**. Specifically, the adhesive force (JIS Z 0237 (2000)) of the lower high adhesive layer **34B** with respect to the resin layer **36** is 10 N/25 mm or greater. Accordingly, when the resin layer **36** is released along the first slits **37**, the base layer **35** and the metal layer **32** that are firmly bonded to the resin layer **36** are released together with the resin layer **36**, and the resin layer **36**, the base layer **35**, and the metal layer **32** are bent along the half cutting portion **39**. A cutting line of perforations may be formed in the base layer **35** to overlap the second slit **33** in a plan view such that the base layer **35** can be broken easily.

(31) As illustrated in FIG. **8**, the IC **40** is an IC chip having a rectangular parallelepiped shape. The ICs **40** (ten ICs in this embodiment) are mounted and one IC **40** is mounted corresponding to each second slit **33** (each storing portion **12**). The IC **40** is mounted on the metal layer **32** so as to cover

a portion of the second slit **33**. A terminal of each IC **40** is connected to a wire that is configured as the antenna formed in the metal layer **32**. In this embodiment, the IC **40** is disposed to cover a portion of the second slit **33** closer to one end portion **33A** (an end portion closer to the outer edge **32A**) of the second slit **33**; however, the IC **40** may be disposed to cover other portion of the second slit **33**. With the second slit **33** having a smaller length, the IC **40** may be disposed to entirely cover the second slit **33**.

(32) According to the PTP **110** having the above configuration, the opening detection function can be achieved in addition to the child-resistant function similar to the first embodiment. More specifically, with the resin layer **36** and the metal layer **32** being released along the first slits **37**, the terminal of the IC **40** is separated from the metal layer **32** and the IC **40** is disconnected from the antenna wire in the metal layer **32** and becomes in a non-conductive state and no communication is possible between the IC **40** and the external communication device. Accordingly, the external communication can determine whether the releasing is performed or not and the opening is detected or not by determining whether or not a predetermined signal is received.

(33) The resin layer **36** is disposed between the cover **14** (the aluminum foil **16**) of the PTP body portion **110** and each of the metal layer **32** (the antenna wire layer) and the ICs **40** that are mounted on the metal layer **32**. According to such a configuration, electromagnetic interference that is caused between them when the communication is performed for the opening detection is suppressed by the insulation effects of the resin layer **36**. As a result, the communication performance is less likely to be lowered and the opening detection can be performed properly. Therefore, with the PTP **110**, the opening detection function having excellent communication performance can be achieved.

Other Embodiments

(34) The present invention is not limited to the embodiments described above with reference to the drawings. For example, the following embodiments may be included in the technical scope of the present invention.

(35) (1) The shape of the protection sheets **30A**, **130A** and the number and the positions of the first slits **37** and the second slits **33** may be altered as appropriate according to the shape of the PTP body portion **11** to which the protection sheet is attached and the number and the positions of the storing portions **12**. For example, for PTP body portions **211**, **311** illustrated in FIGS. **10** and **11**, protection sheets **230A**, **330A** illustrated in FIGS. **12** and **13** may be used, respectively.

(36) (2) The planar shape of the first slit **37** may not be necessarily the I-shape but may be any shapes such as an L-shape and a J-shape as long as the first slit **37** extends until the one end **37A** is at the outer edge **36A** of the resin layer **36** and such that the first slit **37** extends from the one end **37A** and partially along the outer edge (opening edge of the opening **12A**) of the storing portion **12**. With such a shape, after releasing the protection sheet **30A**, **130A**, **230A**, **330A** from the one end **37A** side and along the first slits **37**, a user can press and break the cover **14** that covers the opening **12A** of the storing portion **12**.

(37) (3) A planar size of the protection sheet **30A**, **130A**, **230A**, **330A** may be greater than that of the PTP body portion **11**, **211**, **311**.

(38) (4) The PTP **10**, **110** may be configured to be separated into pieces corresponding to the respective storing portions **12**.

EXPLANATION OF SYMBOLS

(39) **10**, **110**: PTP (press through package), **11**, **211**, **311**: PTP body portion, **12**: storing portion, **13**: sheet-shaped container, **14**: cover, **30**, **30A**, **130A**, **230A**, **330A**: protection sheet, **32**: metal layer, **36**: resin layer, **37**: first slit (slit), **38**: low adhesive layer (first adhesive layer), **39**: half cutting portion (cut portion), **40**: IC, **50**: release paper

Claims

1. A protection sheet to be attached to a press through package that includes a sheet-shaped container including storing portions and a cover that seals the storing portions, the protection sheet comprising: a resin layer being releasable from the press through package and including slits from which the resin layer starts to be released; and a first adhesive layer bonded to the resin layer such that a surface of the first adhesive layer that is an opposite side from the resin layer is to be bonded to the cover, wherein every two slits of the slits are provided for each of the storing portions so as to be opposite each other and sandwich a portion of the resin layer overlapping a corresponding storing portion in a plan view, wherein each of the slits has an elongated shape and extends from a portion of an outer edge of the resin layer in a direction that crosses the outer edge, and wherein a cut portion is on a surface of the resin layer opposite the first adhesive layer and the cut portion extends to connect extending ends of the every two slits.
 2. The protection sheet according to claim 1, wherein the resin layer has a rectangular planar shape as a whole, and each of the slits extends from a portion of a long side included in the outer edge along a short-side direction.
 3. The protection sheet according to claim 1, wherein the resin layer has a rectangular planar shape as a whole, and each of the slits extends from a portion of a short side included in the outer edge along a long-side direction.
 4. The protection sheet according to claim 1, wherein the first adhesive layer is an adhesive agent or an adhesive tape that includes acryl-based material, urethane-based material, silicon-based material, or a rubber-based material.
 5. The protection sheet according to claim 1, further comprising: a metal layer disposed on an opposite side from the first adhesive layer with respect to the resin layer; and an IC mounted on the metal layer to overlap at least a portion of one of the slits in a plan view, the IC being for communicating with an external device.
 6. The protection sheet according to claim 1, further comprising a release paper that is disposed on an opposite side from the resin layer with respect to the first adhesive layer and to be released from the first adhesive layer before being attached to the cover such that the first adhesive layer is uncovered.
 7. A press through package comprising the protection sheet according to claim 1.
 8. The press through package according to claim 7, wherein the protection sheet has a planar shape and a planar size that are substantially same as those of the sheet-shaped container and the cover.
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